

Phase 2 Documentation – Fabulari Chat System

3813ICT Full Stack Development

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GitHub: <https://github.com/HadiAlk4/Chat-App-System>

October 2026

1 Overview

Fabulari is a MEAN-stack chat application: MongoDB (native Node driver), Express 5, Angular 22 (standalone components, PaperCSS) and Node.js, with Socket.io for real-time chat and live request queues. Users chat in rooms that belong to groups. Three roles, Super Admin, Group Admin (GA) and User, control what each person can see and do. Phase 1 kept most data in a JSON file or in mock arrays. In Phase 2 every screen reads and writes MongoDB through the API.

Running it: start MongoDB on localhost:27017; cd backend && npm start (port 3000); cd frontend && npm start (port 4200).

2 Specifications and Requirements

Requirements come from the brief, the client meeting (Week 2) and the published clarifications. Status reflects the code on `main`.

Area	Requirement	Status
Stack	MEAN stack. MongoDB replaces <code>fakeData.json</code> . Socket.io handles real-time events.	Done
Auth	Basic email and password login. Passwords are hashed with bcrypt. OAuth is not used.	Done
Password rule	At least 8 characters, alphanumeric only, at least one uppercase letter. Checked in the browser before sending and again on the server, on signup and password change. Errors show inline in the PaperCSS alert style.	Done
Bootstrap	The first account registered becomes the only Super Admin. Signup shows a note when the database has no users.	Done
Super Admin	Handles requests only. Cannot chat, read chat history, join or propose groups. Creates a group only by approving a user proposal, and the proposer becomes the first GA. Cannot delete or ban themselves.	Done
Global ban	Approving an account deletion removes the user and all their messages, and adds their email to a permanent ban list. Banned emails cannot log in or register again.	Done
Audit log	Every Super Admin decision is stored with a timestamp. A dedicated page filters by action type and date range.	Done
Groups	Unique group names. Any user can view all groups and request to join. No limit on how many groups a user joins or administers.	Done
Age limit	Min age applies to every room in the group. Users under it cannot request to join. Raising it removes members who are too young, but cannot remove the last GA.	Done
GA settings	Edit name, description (≤ 250 chars), min age and theme without Super Admin approval. A confirm dialog lists the changes first.	Done

Area	Requirement	Status
Rooms	GA creates, renames and deletes rooms. Users propose rooms and the GA approves or rejects them. Zero to unlimited rooms per group.	Done
Members	GA can promote members to GA. A group always keeps at least one GA. GA sees the allowed and banned member lists.	Done
Group bans	Bans are permanent and there is no un-ban. Users can ask a GA to ban someone. A ban on a GA goes to the Super Admin instead.	Done
Step down	A GA cannot step down if they are the only GA or have pending Super Admin requests.	Done
Group deletion	A GA requests deletion and the Super Admin approves it. The group and its data are removed, and the GA is demoted if they administer no other group.	Done
Requests	Rejections need a reason. Pending requests cannot be cancelled. Users see pending and rejected requests in Request History.	Done
Messaging	Single-channel, single-thread. Text plus PNG, JPEG and GIF images up to 2MB. No external links. Shows sender name and timestamp. Users can delete their own messages.	Done
History	Only the last 5 messages are loaded when entering a room. Live messages are not capped while the user stays. A new room starts blank.	Done
Real-time	Join and leave toasts (closable with X), live participant sidebar, GA badge in the sidebar, and a toast when a join request is accepted or denied.	Done
Cascade delete	When an account deletion is approved, that user's messages disappear from every open chat straight away.	Done
Theme	The group's theme (light/dark) applies to its chat rooms. A user can override it with a personal theme that applies across all groups.	Done
Profile	Username (changeable), email (fixed, unique ID), password change with current password, DOB/age, role, profile picture (≤ 2 MB), memberships. Profiles are private.	Done
Responsive	Desktop first and usable at tablet width (PaperCSS grid).	Done
HTTPS	Messages served over HTTPS.	Not done

Known limitations. The server runs on HTTP. The API trusts the username sent by the client and has no session token, so role checks happen in the Angular guard and in selected routes. The upload filter also accepts WebP, which the spec does not list.

3 Server-side API

Base URL `http://localhost:3000`. Bodies are JSON (uploads use `multipart/form-data`). Most responses have the form `{ ok, message, ...data }`. Validation failures return `ok:false` with a 400, 403 or 404 status. Server errors return 500. GET list endpoints return an array.

3.1 REST endpoints

Method	Endpoint	Input	Behaviour
<i>Authentication and users</i>			
POST	<code>/api/auth</code>	email, password	bcrypt compare. Blocks banned emails. Returns the user without the password.
GET	<code>/api/signup/first-user</code>	—	<code>{ firstUser }</code> . True when no users exist yet.
POST	<code>/api/signup</code>	username, email, password, dob, age	Checks the password rule, the ban list and duplicate emails. The first user becomes super-admin .

Method	Endpoint	Input	Behaviour
PATCH	/api/users/username	email, newUsername	Must be unique. Renames the user in groups, messages and requests.
PATCH	/api/users/password	email, currentPassword, newPassword	Checks the current password, then applies the rule and stores the new hash.
PATCH	/api/users/theme	email, isDarkMode, usePersonalTheme	Saves the personal chat theme override.
<i>Groups, rooms and members</i>			
GET	/api/groups	–	All groups.
GET	/api/groups/user/:username	–	Groups where the user is a member or admin.
GET	/api/groups/:groupName	–	One group, 404 if missing.
PATCH	/api/groups/:groupName	groupName, groupDescription, minAge, themeColor, username	GA only (403 otherwise). Unique name, description ≤250 chars. Removes members under the new min age. A rename updates every collection.
POST	/api/groups/:groupName/leave	username	Leave the group. The only GA cannot leave.
POST	/api/groups/:groupName/rooms/direct	roomName	GA adds a room. Names are unique within the group, ignoring case.
PATCH	/api/groups/:groupName/rooms/rename	oldName, newName	Rename a room.
DELETE	/api/groups/:groupName/rooms/:roomName	–	Delete a room.
POST	/api/rooms	groupName, roomName	Older direct add from Phase 1.
PATCH	/api/groups/:g/members/:u/promote	–	Add the member to admins .
POST	/api/groups/:g/members/:u/remove	–	Remove the member. The only GA cannot be removed.
POST	/api/groups/:g/members/:u/ban	–	Permanent group ban (bannedMembers). The only GA cannot be banned.
POST	/api/groups/:g/admins/:u/step-down	–	Blocked if they are the only GA or have pending Super Admin requests. Demotes their role if they administer no other group.
<i>Request queues (each has create, list, approve, reject; reject needs reason)</i>			
POST, GET	/api/group-requests	groupName, description, minAge, themeColor, creator	Group proposal to the Super Admin. The name must be free. The Super Admin cannot propose.
PATCH	/api/group-requests/:id/approve reject	performedBy, reason	Approve creates the group with a Main Room and the proposer as GA. Both are audited.
POST, GET	/api/join-requests	groupName, username	Checks age, ban and duplicate requests. The Super Admin cannot join.
PATCH	/api/join-requests/:id/approve reject	reason	Approve adds the user to members .
POST, GET	/api/room-requests	groupName, roomName, username	Members only. The room name must be free.
PATCH	/api/room-requests/:id/approve reject	reason	Approve adds the room.
POST, GET	/api/group-ban-requests	groupName, targetUsername, requestedBy	Routed to the GA, or to the Super Admin if the target is a GA. Filter with destination .

Method	Endpoint	Input	Behaviour
PATCH	/api/group-ban-requests/:id/approve reject	reason	Approve applies the permanent group ban.
POST, GET	/api/group-deletion-requests	groupName, requestedBy, reason	GA only. One pending request per group.
PATCH	/api/group-deletion-requests/:id/approve reject	performedBy, reason	Approve deletes the group, its messages and its requests, and demotes the GA. Audited.
POST, GET	/api/account-deletion-requests	username	The user asks to delete their account. Not allowed for the Super Admin.
PATCH	/api/account-deletion-requests/:id/approve reject	performedBy, reason	Approve deletes the user and their messages and bans the email. Audited as a global ban.
GET	/api/banned-emails	–	Permanently banned emails, newest first.
<i>Chat, uploads and audit</i>			
GET	/api/messages/:groupName/:roomName	query: username	Last 5 messages, oldest first. Returns 403 for the Super Admin.
DELETE	/api/messages/:id	–	Delete a message.
POST	/api/upload/avatar	form: image, username	Image ≤ 2 MB. Sets profilePictureUrl .
POST	/api/upload/chat	form: image	Image ≤ 2 MB. Returns fileUrl (served from /uploads).
GET	/api/audit-logs	query: action, startDate, endDate	Filtered audit log, newest first.

3.2 Socket.io events

Room channels are named `groupName:roomName`. The server ignores chat events from unknown users and from the Super Admin, and drops messages that contain `http(s)://` or `www..`

Direction	Event	Payload and effect
client→server	join-room, leave-room	{groupName, roomName, username}. Joins or leaves the channel and updates who is present.
client→server	send-message	{groupName, roomName, senderUserName, content, imageUrl}. Saves to messages , then broadcasts it.
client→server	delete-message	{groupName, roomName, messageId}. Deletes from MongoDB, then broadcasts.
server→room	new-message, message-deleted	The saved message, or {messageId}.
server→room	user-joined, user-left, room-users	Trigger toasts and the live participant list.
server→all	<queue>-created, <queue>-resolved	For group, join, room, group-ban, group-deletion and account-deletion requests. Queues and Request History refresh live. join-request-resolved triggers the accept/deny toast. account-deletion-request-resolved removes the deleted user's messages from open chats.

4 Angular Architecture

Standalone components with routing in `app.routes.ts`. The logged-in user is kept in `sessionStorage`, so each browser tab has its own login. `authGuard` protects every route except login and signup. It checks `expectedRole` for Super Admin pages and sends a Super Admin away from `chat`, `my-memberships` and `group-settings`.

4.1 Components

Component	Route	Responsibility
Login	/login	Posts credentials to <code>/api/auth</code> , stores the user, and routes by role (Super Admin dashboard or dashboard). Shows API errors, including the banned message.
Signup	/signup	Registration with age from DOB and an optional profile doodle. Shows a note when the next account will be the Super Admin.
Dashboard	/dashboard	Lists and searches groups, proposes a group (modal), and requests to join. Groups the user is already in are hidden. Both actions are hidden for the Super Admin.
MyMemberships	/my-memberships	The user's groups, filtered by name and role. Opens chat, settings, or leave.
GroupSettings	/group-settings	GA-only tabs. <i>General</i> : edit and save with a confirm dialog. <i>Rooms</i> : add, rename, delete. <i>Members</i> : promote, ban, banned list. <i>Requests</i> : join, room and ban queues. Also step down and request deletion. Updates live over sockets.
RequestHistory	/request-history	The user's pending and rejected requests with reasons. Tabs to propose a room and to request a ban.
Chat	/chat	Room list, last-5 history plus live messages, image attachments, delete own message, toasts, participant sidebar with GA badge, and the group or personal theme.
UserProfileSettings	/user-profile-settings	Change username, password and theme. Upload a profile picture. Request account deletion.
SuperAdminDashboard	/super-admin-dashboard	Live queues for group proposals, GA ban requests, group deletions and account deletions, plus the banned-email list. Every rejection needs a reason.
SuperAdminAuditLog	/super-admin-audit-log	Audit table filtered by action type and date range.

4.2 Services

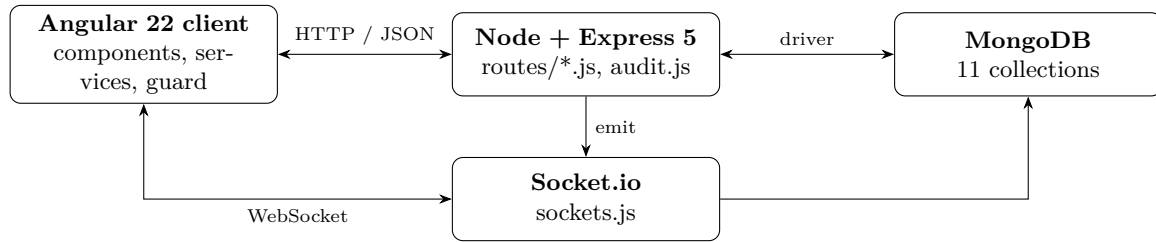
- **AuthService**: get, set and clear the session user; logout.
- **GroupService**: all group, room and member endpoints, plus every request queue (group, join, room, ban, deletion).
- **ChatService**: room history over HTTP. Emits and listens to chat socket events (`join-room`, `send-message`, `new-message`, `room-users`, and so on).
- **SocketService**: typed observables for every `*-created` and `*-resolved` request event.
- **AccountService**: account deletion requests, banned emails, and username, password and theme updates.
- **UploadService**: avatar and chat image uploads (`FormData`).
- **AuditService**: audit log query with filters.

4.3 Models (TypeScript interfaces)

`Group`, `GroupRequest`, `JoinRequest`, `RoomRequest`, `GroupBanRequest`, `GroupDeletionRequest`, `AccountDeletionRequest`, `BannedEmail`, `AuditLog`, `ChatMessage`. They mirror the MongoDB documents in Section 5.2. Request models share a `status` of `pending`, `approved` or `rejected`, plus `createdAt`, `reviewedAt` and `rejectionReason`.

5 Design Documents

5.1 System architecture



`app.js` builds the Express app (`createApp(io)`). `index.js` attaches the HTTP server and Socket.io. Because of this split, tests can import the app without starting a server, using a stub `io`. Business rules are pulled out into small, pure helpers in `backend/lib/` (password policy, age check, link check, last-5 window).

5.2 Data structures (MongoDB collections)

Collection	Fields
<code>users</code>	username, email (unique ID), password (bcrypt), dob, age, role (<code>super-admin group-admin user</code>), valid, isDarkMode, usePersonalTheme, profilePictureUrl
<code>groups</code>	groupName (unique), groupDescription, minAge, themeColor (<code>light dark</code>), admins[], members[], rooms[], bannedMembers[]
<code>messages</code>	groupName, roomName, senderUserName, content, imageUrl, timestamp
<code>groupRequests</code>	groupName, groupDescription, minAge, themeColor, creatorUserName, creatorEmail + request fields
<code>joinRequests</code>	groupName, username + request fields
<code>roomRequests</code>	groupName, roomName, username + request fields
<code>groupBanRequests</code>	groupName, targetUsername, targetEmail, targetRole, requestedBy, requestedByRole, destination (<code>group-admin super-admin</code>) + request fields
<code>groupDeletionRequests</code>	groupName, requestedBy, requestedByRole, reason + request fields
<code>accountDeletionRequests</code>	username, email, role + request fields
<code>bannedEmails</code>	email, username, bannedAt, reason
<code>auditLogs</code>	timeStamp, actionPerformed, target, performedBy (insert only)

Request fields are status, createdAt, reviewedAt and rejectionReason. Groups store room names and usernames as arrays instead of references, so a single read loads a whole group. When a user or group is renamed, the server updates the name in every collection.

5.3 Storyboards

The Phase 1 storyboards (desktop and tablet) still describe the layout, so they were kept. They are in Appendix A and in `assignment_material/Fabulari Design/`. Changes made in Phase 2:

- The Group Settings *General* tab is a real form and asks for confirmation before saving.
- Request History has a *Request Ban* tab.
- The Super Admin dashboard has live GA ban and group deletion queues.
- Chat has image attachments and a delete button on your own messages.
- Every Super Admin rejection asks for a reason.

6 Testing

6.1 Tools and methodology

- **Backend unit tests**, Mocha with Node `assert` (`npm run unitTest`). Test the pure helpers in `backend/lib`.

- **Backend integration tests**, Mocha, Chai and chai-http (`npm test`). Each test calls `createApp()` against a separate `chat-app-test` database. The database is wiped before and after every test, so no test depends on another. Each endpoint has at least one success case and one failure case.
- **Angular unit tests**, Vitest with TestBed via `ng test`. Services are replaced with test doubles, so components are tested without a backend.
- **End-to-end tests**, Cypress (`npm run cypress:run`), against the real frontend and a backend started with `MONGO_DB_NAME=chat-app-test`. A `resetDb` task drops the test database before each spec.

Before any tests were written, the logic was moved into testable helpers and the app setup was split from `listen()`. The end-to-end tests found four screens that did not refresh after data loaded (login errors, dashboard, memberships, chat), and these were fixed.

Results (run 1 Oct 2026): all 153 automated tests pass: backend unit 14/14, backend integration 100/100, Angular 30/30 and Cypress 9/9. The full output of each run is in Section 6.3.

6.2 Automated tests

Suite	Target	#	Cases
Unit	<code>isValidPassword</code>	4	accepts valid; rejects too short, no uppercase, symbol
Unit	<code>isUnderMinAge</code>	3	under min; equal to min; missing age not treated as under
Unit	<code>containsExternalLink</code>	3	detects http, www; allows plain text
Unit	<code>latestMessagesOldestFirst</code>	4	keeps latest 5; exactly 5; fewer than 5; oldest-first order
Integration	<code>/api/auth</code> , <code>/api/signup</code>	6	valid login; wrong password; banned login; first user is Super Admin; weak password; banned email signup
Integration	<code>/api/users/*</code>	6	rename; missing name; change password; wrong current password; save theme; missing theme flags
Integration	<code>/api/groups*</code>	11	list all / empty; user groups / none; get one / 404; leave / missing username; save settings and remove under-age members; missing description; non-GA 403
Integration	rooms	8	add / missing name; rename / missing new name; delete / 404; <code>POST /api/rooms</code> add / unknown group
Integration	members	8	promote / 404; remove / only GA; ban / 404; step down / only GA
Integration	group requests	8	create / missing fields; list / empty filter; approve creates group with GA / not pending; reject with reason / no reason
Integration	join requests	8	create / under age; list / empty filter; approve adds member / not pending; reject with reason / no reason
Integration	room requests	8	create / missing fields; list / empty filter; approve adds room / not pending; reject with reason / no reason
Integration	group ban requests	8	create / missing fields; list by destination / empty; approve bans / 404; reject with reason / no reason
Integration	group deletion	8	create / missing reason; list / empty; approve deletes and demotes / 404; reject with reason / no reason
Integration	account deletion, banned emails	10	create / Super Admin blocked; list / empty; approve deletes and bans / 404; reject with reason / no reason; banned list / empty
Integration	<code>/api/messages</code>	5	last 5 oldest first; missing username 400; Super Admin 403; delete / 404
Integration	<code>/api/upload/*</code>	4	avatar PNG / no file; chat PNG / no file
Integration	<code>/api/audit-logs</code>	2	returns logs; filters by action
Angular	App	2	creates root; renders router outlet
Angular	authGuard	3	logged out to login; user blocked from Super Admin page; Super Admin blocked from chat
Angular	Login	3	empty fields no HTTP; user to dashboard; Super Admin to Super Admin dashboard
Angular	Signup	6	age for past, empty and future DOB; empty form not posted; weak password not posted; invalid email not posted
Angular	Dashboard	2	proposal sent via service; joined groups hidden

Suite	Target	#	Cases
Angular	Chat	3	link not sent; plain text sent; history shows sender name
Angular	GroupSettings	4	form filled from group; confirm before raising min age; rejection needs reason; non-GA redirected
Angular	RequestHistory, MyMemberships	2	renders a join request; renders a membership
Angular	SuperAdminDashboard	1	renders pending proposals
Angular	SuperAdminAuditLog	2	renders logs; applies action and date filters
Angular	UserProfileSettings	2	shows error on wrong current password; weak new password not sent
E2E	login.cy.ts	3	logs in and fills dashboard; sends credentials to API; error on empty submit
E2E	access.cy.ts	3	logged-out redirect; logged-in reaches dashboard; bad credentials stay on login
E2E	admin-group.cy.ts	1	after Super Admin approval the group leaves the creator's join list and appears in My Memberships
E2E	chat.cy.ts	2	two users: other member's message shows their name; link message not shown

6.3 Test run output

Output from the four test commands, pasted as printed. Colour codes, Node/npm warnings and a Cypress start-up warning about clearing old screenshots are removed, and tick, cross and box-drawing characters are replaced with ASCII so the PDF font can display them.

Backend unit tests (cd backend && npm run unitTest)

```
> chat-app-system@1.0.0 unitTest
> mocha unitTest

isUnderMinAge
  under the minimum
    ok returns true when the age is below the group minimum
  equal to the minimum
    ok returns false when the age matches the group minimum
  missing age
    ok does not treat a missing age as under the minimum

containsExternalLink
  http link
    ok detects an http URL
  www link
    ok detects a www host
  plain text
    ok allows text that is not a link

latestMessagesOldestFirst
  more than five
    ok keeps only the latest five messages
  exactly five
    ok returns all five messages
  fewer than five
    ok returns every message when the room has fewer than five
  oldest-first order
    ok returns the window with the oldest message first

isValidPassword
  valid password
    ok accepts 8 alphanumeric characters with an uppercase letter
  too short
    ok rejects a password shorter than 8 characters
  missing uppercase
    ok rejects a long password with no uppercase letter
  symbol rejected
    ok rejects a password that contains a symbol

14 passing (6ms)
```

Backend integration tests (cd backend && npm test)

```
> chat-app-system@1.0.0 test
> MONGO_DB_NAME=chat-app-test mocha --require ./integrationTest/hooks.js --timeout 20000 "integrationTest/*.test.js"

account deletion routes
  POST /api/account-deletion-requests
    ok queues a deletion request (99ms)
    ok rejects a super admin deletion request
  GET /api/account-deletion-requests
    ok returns pending requests
    ok returns an empty list when the status filter matches nothing
  PATCH /api/account-deletion-requests/:id/approve
    ok deletes the user and bans the email (78ms)
    ok returns 404 when the request is not pending
  PATCH /api/account-deletion-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason
  GET /api/banned-emails
    ok returns banned emails
    ok returns an empty list when nobody is banned

audit routes
  GET /api/audit-logs
    ok returns stored audit logs
    ok filters logs by action type

auth routes
  POST /api/auth
    ok logs in with a valid email and password
    ok rejects a wrong password
    ok tells a permanently banned account it cannot log in
  POST /api/signup
    ok creates the first account as super admin (114ms)
    ok rejects a password without an uppercase letter
    ok rejects a permanently banned email

chat routes
  GET /api/messages/:groupName/:roomName
    ok returns the latest five messages oldest first (46ms)
    ok rejects a history request with no username
    ok forbids a super admin from reading chat history
  DELETE /api/messages/:id
    ok deletes a stored message
    ok returns 404 when the message does not exist

group ban request routes
  POST /api/group-ban-requests
    ok queues a ban request against another member (84ms)
    ok rejects a request that is missing fields
  GET /api/group-ban-requests
    ok returns pending requests for a destination
    ok returns an empty list when the destination filter matches nothing
  PATCH /api/group-ban-requests/:id/approve
    ok bans the target member (41ms)
    ok returns 404 when the request is not pending
  PATCH /api/group-ban-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group deletion routes
  POST /api/group-deletion-requests
    ok queues a deletion request from a group admin (45ms)
    ok rejects a request that is missing a reason
  GET /api/group-deletion-requests
    ok returns pending requests
    ok returns an empty list when the status filter matches nothing
  PATCH /api/group-deletion-requests/:id/approve
    ok deletes the group and demotes the requesting admin
    ok returns 404 when the request is not pending
  PATCH /api/group-deletion-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group request routes
  POST /api/group-requests
    ok queues a proposal (46ms)
    ok rejects a proposal that is missing fields
  GET /api/group-requests
    ok returns pending proposals
    ok returns an empty list when the status filter matches nothing
  PATCH /api/group-requests/:id/approve
    ok creates the group and assigns the proposer as admin
    ok rejects an id that is not a pending proposal
  PATCH /api/group-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group routes
  GET /api/groups
```

```

    ok returns every group
    ok returns an empty list when no groups exist
GET /api/groups/user/:username
    ok returns groups the user belongs to
    ok returns an empty list for a user who belongs to no group
GET /api/groups/:groupName
    ok returns one group
    ok returns 404 when the group does not exist
POST /api/groups/:groupName/leave
    ok removes a member who is not the sole admin
    ok rejects a leave request with no username
PATCH /api/groups/:groupName
    ok saves settings and removes members under the new minimum age
    ok rejects a description that is missing
    ok rejects a member who is not a Group Admin

join request routes
POST /api/join-requests
    ok creates a pending join request
    ok rejects a user who is under the group minimum age
GET /api/join-requests
    ok returns pending requests for a group
    ok returns an empty list when the status filter matches nothing
PATCH /api/join-requests/:id/approve
    ok adds the requester to the group
    ok rejects an id that is not a pending request
PATCH /api/join-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

member routes
PATCH /api/groups/:groupName/members/:username/promote
    ok adds a member to the admin list
    ok returns 404 when the user is not in the group
POST /api/groups/:groupName/members/:username/remove
    ok removes a member who is not the sole admin
    ok rejects removal of the sole administrator
POST /api/groups/:groupName/members/:username/ban
    ok permanently bans a member
    ok returns 404 when the user is not in the group
POST /api/groups/:groupName/admins/:username/step-down
    ok steps down when another group admin remains
    ok rejects a step-down by the sole group admin

room request routes
POST /api/room-requests
    ok creates a pending room proposal
    ok rejects a proposal that is missing fields
GET /api/room-requests
    ok returns pending proposals for a group
    ok returns an empty list when the status filter matches nothing
PATCH /api/room-requests/:id/approve
    ok adds the proposed room
    ok rejects an id that is not a pending proposal
PATCH /api/room-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

room routes
POST /api/groups/:groupName/rooms/direct
    ok adds a room
    ok rejects a missing room name
PATCH /api/groups/:groupName/rooms/rename
    ok renames a room
    ok rejects a rename that omits the new name
DELETE /api/groups/:groupName/rooms/:roomName
    ok deletes a room
    ok returns 404 when the room does not exist
POST /api/rooms
    ok adds a room to an existing group
    ok rejects a room for a group that does not exist

upload routes
POST /api/upload/avatar
    ok stores a small PNG as the profile picture
    ok rejects a request with no file
POST /api/upload/chat
    ok accepts a small PNG
    ok rejects a request with no file

user routes
PATCH /api/users/username
    ok renames the account
    ok rejects a missing new username
PATCH /api/users/password
    ok changes the password when the current password matches (262ms)
    ok rejects a wrong current password
PATCH /api/users/theme
    ok saves the chat theme flags
    ok rejects a theme update that omits the boolean flags

```

Database connection closed

100 passing (2s)

Angular unit tests (cd frontend && ng test --watch=false --reporters=verbose)

```
> Building...
ok Building...
Initial chunk files      | Names      | Raw size
init-testbed.js         | init-testbed | 346.04 kB |
spec-app-group-settings-group-settings.js | spec-app-group-settings-group-settings | 87.05 kB |
spec-app-request-history-request-history.js | spec-app-request-history-request-history | 72.27 kB |
styles.css              | styles      | 66.10 kB |
spec-app-chat-chat.js   | spec-app-chat-chat | 57.30 kB |
spec-app-super-admin-dashboard-super-admin-dashboard.js | spec-app-super-admin-dashboard-super-admin-dashboard | 51.99 kB |
spec-app-user-profile-settings-user-profile-settings.js | spec-app-user-profile-settings-user-profile-settings | 39.74 kB |
spec-app-dashboard-dashboard.js | spec-app-dashboard-dashboard | 33.15 kB |
spec-app-my-memberships-my-memberships.js | spec-app-my-memberships-my-memberships | 24.00 kB |
spec-app-signup-signup.js | spec-app-signup-signup | 22.81 kB |
spec-app-super-admin-audit-log-super-admin-audit-log.js | spec-app-super-admin-audit-log-super-admin-audit-log | 18.54 kB |
spec-app-login-login.js | spec-app-login-login | 12.28 kB |
spec-app-guards-auth-guard.js | spec-app-guards-auth-guard | 4.16 kB |
spec-app-app.js         | spec-app-app | 1.93 kB |
vitest-mock-patch.js    | vitest-mock-patch | 988 bytes |
setup-test-setup.js     | setup-test-setup | 777 bytes |

| Initial total | 839.12 kB

Application bundle generation complete. [1.917 seconds] - 2026-10-01T00:01:21.705Z

RUN v4.1.11 /Users/abdalhadiakhub/Desktop/ChatApp System/Chat-App-System/frontend

stderr | src/app/chat/chat.spec.ts
NG0912: Component ID generation collision detected. Components '_Chat' and '_Chat' with selector 'app-chat' generated the same
↳ component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to force a
↳ different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/login/login.spec.ts
NG0912: Component ID generation collision detected. Components '_Login' and '_Login' with selector 'app-login' generated the
↳ same component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to
↳ force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/group-settings/group-settings.spec.ts
NG0912: Component ID generation collision detected. Components '_GroupSettings' and '_GroupSettings' with selector
↳ 'app-group-settings' generated the same component ID. To fix this, you can change the selector of one of those components or
↳ add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/signup/signup.spec.ts
NG0912: Component ID generation collision detected. Components '_Signup' and '_Signup' with selector 'app-signup' generated the
↳ same component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to
↳ force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/super-admin-dashboard/super-admin-dashboard.spec.ts
NG0912: Component ID generation collision detected. Components '_SuperAdminDashboard' and '_SuperAdminDashboard' with selector
↳ 'app-super-admin-dashboard' generated the same component ID. To fix this, you can change the selector of one of those
↳ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/super-admin-audit-log/super-admin-audit-log.spec.ts
NG0912: Component ID generation collision detected. Components '_SuperAdminAuditLog' and '_SuperAdminAuditLog' with selector
↳ 'app-super-admin-audit-log' generated the same component ID. To fix this, you can change the selector of one of those
↳ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/request-history/request-history.spec.ts
NG0912: Component ID generation collision detected. Components '_RequestHistory' and '_RequestHistory' with selector
↳ 'app-request-history' generated the same component ID. To fix this, you can change the selector of one of those components
↳ or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/login/login.spec.ts > Login > shows an error and does not call HTTP when the fields are empty 100ms
ok |frontend| src/app/login/login.spec.ts > Login > navigates a regular user to the dashboard after a successful login 12ms
ok |frontend| src/app/login/login.spec.ts > Login > navigates a super admin to the super admin dashboard 49ms
stderr | src/app/dashboard/dashboard.spec.ts
NG0912: Component ID generation collision detected. Components '_Dashboard' and '_Dashboard' with selector 'app-dashboard'
↳ generated the same component ID. To fix this, you can change the selector of one of those components or add an extra host
↳ attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/super-admin-dashboard/super-admin-dashboard.spec.ts > SuperAdminDashboard > renders pending group
↳ proposals from the group service 144ms
stderr | src/app/user-profile-settings/user-profile-settings.spec.ts
NG0912: Component ID generation collision detected. Components '_UserProfileSettings' and '_UserProfileSettings' with selector
↳ 'app-user-profile-settings' generated the same component ID. To fix this, you can change the selector of one of those
↳ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/dashboard/dashboard.spec.ts > Dashboard > sends a named proposal through the group service 44ms
ok |frontend| src/app/dashboard/dashboard.spec.ts > Dashboard > hides groups the user already belongs to from the join list
↳ 21ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > fills the form from the loaded group 165ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > asks for confirmation before saving a higher
↳ minimum age 64ms
```

```

ok |frontend| src/app/chat/chat.spec.ts > Chat > does not send a message that contains an external link 280ms
ok |frontend| src/app/super-admin-audit-log/super-admin-audit-log.spec.ts > SuperAdminAuditLog > renders the mocked audit log
↳ 176ms
ok |frontend| src/app/super-admin-audit-log/super-admin-audit-log.spec.ts > SuperAdminAuditLog > applies the action and date
↳ filters to the mocked list 49ms
ok |frontend| src/app/chat/chat.spec.ts > Chat > sends a plain text message 52ms
ok |frontend| src/app/chat/chat.spec.ts > Chat > renders the room history with the sender display name 14ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > does not call the API when a join rejection has
↳ no reason 48ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings access > sends a regular member away from group
↳ settings 35ms
ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a logged-out visitor to the login page 2ms
ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a user away from a super admin page 1ms
ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a super admin away from chat 1ms
stderr | src/app/my-memberships/my-memberships.spec.ts
NG0912: Component ID generation collision detected. Components '_MyMemberships' and '_MyMemberships' with selector
↳ 'app-my-memberships' generated the same component ID. To fix this, you can change the selector of one of those components or
↳ add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/request-history/request-history.spec.ts > RequestHistory > renders the mocked join request 204ms
ok |frontend| src/app/app.spec.ts > App > creates the root component 3ms
ok |frontend| src/app/app.spec.ts > App > renders the router outlet 3ms
ok |frontend| src/app/my-memberships/my-memberships.spec.ts > MyMemberships > renders the mocked membership 18ms
ok |frontend| src/app/user-profile-settings/user-profile-settings.spec.ts > UserProfileSettings > shows the service error when
↳ the current password is wrong 80ms
ok |frontend| src/app/user-profile-settings/user-profile-settings.spec.ts > UserProfileSettings > does not call the API when
↳ the new password breaks the password rule 15ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > calculates a positive age for a past date of birth 377ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > returns -1 when the date of birth is empty 17ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > returns a negative age for a future date of birth 17ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > does not post when the form is empty 17ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > does not post a password that breaks the password rule 18ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > does not post an invalid email address 15ms

Test Files 12 passed (12)
Tests 30 passed (30)
Start at 10:01:21
Duration 3.08s (transform 1.74s, setup 6.54s, import 823ms, tests 2.06s, environment 15.34s)

```

End-to-end tests (cd frontend && npx cypress run)

```

=====

(Run Starting)

+-----+
| Cypress:      16.1.0                               |
| Browser:      Electron 146 (headless) (deprecated) |
| Node Version: v26.8.2 (/opt/homebrew/Cellar/node/26.8.2/bin/node) |
| Specs:         4 found (access.cy.ts, admin-group.cy.ts, chat.cy.ts, login.cy.ts) |
| Searched:    cypress/e2e/**/*.cy.{js,jsx,ts,tsx}  |
+-----+

Warning: The Electron browser is deprecated as a test browser and will be removed in a future version of Cypress.

Switch to Chrome or another installed browser to avoid a breaking change when you upgrade.

Read more about supported browsers: https://on.cypress.io/launching-browsers

-----

Running:  access.cy.ts                                     (1 of 4)

Access control
  ok redirects a logged-out visit to the dashboard back to login (1417ms)
  ok lets a logged-in user reach the dashboard (787ms)
  ok keeps a user with bad credentials on the login page (628ms)

3 passing (3s)

(Results)

+-----+
| Tests:        3                                     |
| Passing:      3                                     |
| Failing:      0                                     |
| Pending:      0                                     |
| Skipped:      0                                     |
| Screenshots:  0                                     |
| Video:        false                                 |
| Duration:     2 seconds                             |
| Spec Ran:     access.cy.ts                         |
+-----+

-----

Running:  admin-group.cy.ts                                (2 of 4)

```

Group approval
ok shows a group to its creator once the Super Admin approves it (3030ms)

1 passing (3s)

(Results)

```
+-----+
| Tests:      1 |
| Passing:    1 |
| Failing:    0 |
| Pending:    0 |
| Skipped:    0 |
| Screenshots: 0 |
| Video:      false |
| Duration:   3 seconds |
| Spec Ran:   admin-group.cy.ts |
+-----+
```

Running: chat.cy.ts

(3 of 4)

Chat

ok shows another member's message with their display name (1482ms)
ok does not show a message that contains an external link (1083ms)

2 passing (3s)

(Results)

```
+-----+
| Tests:      2 |
| Passing:    2 |
| Failing:    0 |
| Pending:    0 |
| Skipped:    0 |
| Screenshots: 0 |
| Video:      false |
| Duration:   2 seconds |
| Spec Ran:   chat.cy.ts |
+-----+
```

Running: login.cy.ts

(4 of 4)

Login

ok logs in and fills the dashboard (1141ms)
ok sends the credentials to the auth API (655ms)
ok shows an error when submitted empty (314ms)

3 passing (2s)

(Results)

```
+-----+
| Tests:      3 |
| Passing:    3 |
| Failing:    0 |
| Pending:    0 |
| Skipped:    0 |
| Screenshots: 0 |
| Video:      false |
| Duration:   2 seconds |
| Spec Ran:   login.cy.ts |
+-----+
```

=====
(Run Finished)

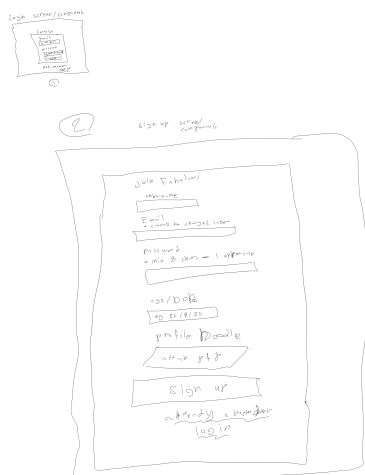
Spec	Tests	Passing	Failing	Pending	Skipped
ok access.cy.ts	00:02	3	3	-	-
ok admin-group.cy.ts	00:03	1	1	-	-
ok chat.cy.ts	00:02	2	2	-	-
ok login.cy.ts	00:02	3	3	-	-
ok All specs passed!	00:10	9	9	-	-

7 Git Workflow

Every feature was built on its own branch and merged into `main` through a GitHub pull request (221 commits, PRs #1 to #52). Each PR description lists what the branch changed. Phase 2 branches, in order:

Branch (PR)	Work
feature/auth-route-guard (#28)	AuthService, route guard with roles, logout on every screen
feature/auth-and-live-groups (#30, #31)	bcrypt and Super Admin bootstrap, Group models and service, dashboard loads live groups
feat/group-proposal-flow (#32, #33)	Group proposal to Super Admin queue; first Socket.io events
feature/group-join-requests (#34, #35)	Join request model, routes, sockets, GA queue, Request History
refactor/backend-mongodb-structure (#36)	Split backend into <code>db.js</code> and <code>routes/*</code> modules
feat/room-proposal (#37, #38)	Room proposals wired to Group Settings
feature/my-memberships (#39)	Memberships from MongoDB
feature/group-settings (#40)	Rooms and Members tabs on live data
feat/chat (#41, #42)	Messages collection, chat sockets, last-5 history, delete, live participants
feature/file-attachments (#43)	Multer uploads for avatars and chat images
account-deletion-bans (#44)	Account deletion queue, permanent email bans
audit-log-mongodb (#45)	Audit log collection and filtered page
user-profile-mdb (#46)	Username, password and theme updates
group-admin-remaining (#47)	Unique names, settings save, age eviction, group bans, ban requests, step down, group deletion
feat/sa-request-only-scope (#48)	Super Admin limited to requests (UI, guard, API, sockets)
feat/chat-toast-cascade-theme (#49)	Join toasts, live cascade delete, link blocking, personal theme
testing (#50)	Testable refactor, then Mocha, Chai, Vitest and Cypress suites
cleanup-bugs (#51, #52)	GA check on settings, confirm dialog, first-user note, per-tab sessions

A Storyboards



Login / Sign up



Dashboard

Current user: _____ | Role: _____

Search by group name: _____ All Roles _____ Filter _____

Title	Rule	Description
View Rooms	Learn	Group Settings

Title	Rule	Description
View Rooms	Learn	

Title	Rule	Description
View Rooms	Learn	Group Settings

Title	Rule	Description
View Rooms	Learn	Group Settings

Title	Rule	Description
View Rooms	Learn	Group Settings

Title	Rule	Description
View Rooms	Learn	Group Settings

Title	Rule	Description
View Rooms	Learn	Group Settings

The diagram illustrates a login system interface with the following components:

- Group Name** (Group Name): A label for the group selection.
- Friday Lab** (Freitag Lab): A label for the specific lab session.
- User:** A label for the user input field.
- Passwort** (Password): A label for the password input field.
- LogIn**: A button to submit the login information.
- Group Members** (Group Members): A list of group members, including:
 - Admin:
 - User:
 - User:
 - Admin:
 - User:
- Online Room Members** (Online Room Members): A list of online room members, including:
 - Join / Leave
 - Group
 - Member
- 2. Text a Message** (2. Text a Message): A label for the messaging section.
- Send**: A button to send the message.
- 2. Text a Message** (2. Text a Message): A label for the messaging section.
- 2. Text a Message** (2. Text a Message): A label for the messaging section.

The diagram illustrates a registration form layout with the following elements:

- Account Settings** (Main Title)
- User name:** (Label) followed by a **First Name** input field.
- Account Details:** (Section Header) followed by:
 - Email Address** input field
 - Birth Date** input field
 - Sign-in Role** input field
 - My Membership** button
- Identify & Preferences** (Section Header) followed by:
 - Name** input field
 - Save** button
 - Dark Mode Theme** (Label) with a **Light** button and a **Dark** button.
 - go to the full screen.** (Text)
- Related Account Options** (Section Header) at the bottom.

Profile Poodle

would to change your pass?

upload a new doodle below (mouse)

Change Data

upload a New doodle.

Security

Current Password

New Password

Confirm New Password

Save New Password

S

Super Admin Audit Log

Name: _____
Role: _____

Super Admin Dashboard

#	Timestamp	Action performed	Target IP
1-	_____	_____	_____
2.	_____	_____	_____
3-	_____	_____	_____
...	_____	_____	_____

log out

15